Projectiles persevere in their motions, so far as they are not retarded by the resistance of the air, or impelled downwards by the force of gravity. A stone or a cannonball, once flung, would fly straight on forever were these two forces absent — shot fast enough horizontally, it would circle the whole Earth. What we see as a falling arc is only the projectile's straight perseverance bent by $\vec{a} = \vec{g} + \frac{\vec{F}_{\mathbf{air}}}{m}$.